GLOBAL SCHOOL OF SCIENCE

First Term Examination - 2080

Class : XI | **Subject: Mathematics (0071)** | **Full Marks: 75**

Time: 3 hrs. | **SET - B** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A' **[11 × 1 = 11]**

Rewrite the best alternative of each question in your answer sheet.

1. The number of the proper subsets of a set A of order n is equal to….

   a) $n-1$ b) $2n-1$ c) 2^n-1 d) n^2-1

2. If $A \cap B = \phi$, then $A \cup \overline{B} =$

   a) A b) B c) $\bar{B}$ d) $A \cap B$

3. $-3 \leq x \leq -1$ is same as

   a) $|x-3| \leq 5$ b) $|x+3| \leq 1$

   c) $|x+2| \leq 1$ d) $|3x-1| \leq 4$

4. Both roots of the quadratic equation $ax^2+bx+c=0$ are zeros, if

   a) $b=0, c=0$ b) $b=0, c\neq 0$

   c) $b\neq 0, c=0$ d) $b\neq 0, c\neq 0$

5. Inverse of a function f(x) exists if it is

   a) injective function b) surjective function

   c) bijective function d) into function

6. Two matrices A and B are inverse to each other if and only if

   a) $AB=BA=0$ b) $AB=BA$

   c) $AB=I, BA=0$ d) $AB=BA=I$

7. If A is a square matrix of order n and A = kB where k is a scalar then |A| =

a) |B| b) k|B| c) k^n|B| d) n|B|

8. If $\begin{vmatrix} 2 & 5 & 4 \\ 0 & x+4 & 0 \\ -3 & 7 & -1 \end{vmatrix} = 0$, then value of x is

a) -2 b) 2 c) -4 d) 0

9. The point of discontinuity of the function $f(x) = \frac{x^2+6x+8}{x^2-5x+6}$ are

a) $x = 1, 2$ b) $x = 2, 3$ c) $x = 1, 3$ d) $x = -2, 3$

10. The value of $\lim_{x \to 3} \frac{|x-3|}{(x-3)}$ is

a) 1 b) -1 c) 0 d) does not exist

11. The value of $\lim_{x \to 0} \frac{\tan x^g}{x}$ is

a) $\frac{200}{\pi}$ b) $\frac{\pi}{200}$ c) $\frac{180^0}{\pi}$ d) 1

Group 'B'

Short Answer Questions. **[8 × 5 = 40]**

12. a) Define intersection of two sets in set builder form.

b) If A, B, C are any three subsets of universal set U, show that:

i) $A \cap (B - C) = (A \cap B) - (A \cap C)$

ii) $A - (B \cap C) = (A - B) \cup (A - c)$ [1+2+2]

13. a) If $x \in \mathbb{R}$ and a is any positive real number, prove that:

$|x| < a \Rightarrow -a < x < a$ [2]

b) Solve the inequality $|2x - 1| \leq 5$ and show the solution graphically. [2+1]

14. a) A function $f: X \to Y$ is defined by $f(x) = \frac{x+3}{x-1}$ with $X = \{-2, -1, 0, 2\}$

and $Y = \{-3, -1, -\frac{1}{3}, 0, 1, 3, 5\}$. Find the range of f. [2]

b) Define symmetric matrix with an example of order 3. If this matrix is A, find $A - A^T$ and comment on the result. [1+2]

15. a) Using the properties of determinant, prove that:

$$\begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \end{vmatrix} = (x+y+z)^3$$ [3]

b) Prove that the roots of the equation $(x-a)(x-b) = k^2$ are real for all values of k. [2]

16. a) If the equation $(1+m^2)x^2 + 2mcx + (c^2 - a^2) = 0$ has equal roots, show that: $c^2 = a^2(1+m^2)$ [2]

b) Write Eular's constant (e) in terms of a limit as $x \to \infty$.

Prove that: $\lim\limits_{x \to 0} \frac{e^x - 1}{x} = 1$ [1+2]

17. Write any three indeterminate forms. Evaluate: $\lim\limits_{x \to \theta} \frac{x\sin\theta - \theta\sin x}{x - \theta}$ [1+4]

18. a) Define the continuity of a function f(x) at a point x = a. If the function defined by $f(x) = \begin{cases} 2x+1 & when \quad x \neq 1 \\ k & when \quad x = 1 \end{cases}$ is continuous at x = 1, then find the value of k. [1+1]

b) A function $f(x)$ is defined as $f(x) = \begin{cases} x^2 - 1 & for\ x < 2 \\ 1 & for\ x = 2 \\ x+1 & for\ x > 2 \end{cases}$

Is the function $f(x)$ continuous at x = 2 ? If not, write the type of discontinuity and redefine the function $f(x)$ so as to make it continuous there. [2+1]

19. a) If $f(x) = \frac{\sqrt{2x} - \sqrt{3x-a}}{\sqrt{x} - \sqrt{a}}$ then

i) Find the value of f (a) and comment the result. [1]

ii) Evaluate: $\lim\limits_{x \to a} f(x)$ [2]

b) Evaluate: $\lim_{x \to y} \frac{\tan x - \tan y}{\sqrt{x}-\sqrt{y}}$ [2]

Group 'C'

Long Answer Questions. **[3 × 8 = 24]**

20. a) What is the contrapositive of the conditional statement $p \Rightarrow q$. Prove that $\sim(p \Rightarrow q)$ and $p \wedge (\sim q)$ are logically equivalent. Write the contrapositive of the statement "If $a = 0$, then $ax^2 + bx + c = 0$ is not a quadratic equation". [1+2+1]

b) If $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 0 & 1 \\ 0 & 3 & -1 \end{pmatrix}$ then, find adjoint of A and determinant of A. Is it possible to find A^{-1}? Write the reason for it. [2+1+1]

21. a) If A=$\begin{bmatrix} 2 & -1 & 3 \\ 0 & 5 & 1 \\ -1 & -3 & -2 \end{bmatrix}$ then,

i) Write the elements $a_{11},\ a_{21}$ and a_{31} of the matrix A. [1]

ii) Find the minors M_{11}, M_{21} and M_{31} of the elements a_{11}, a_{21} and a_{31}. [2]

iii) Find the cofactors A_{11}, A_{21} and A_{31} of the elements a_{11}, a_{21} and a_{31}. [1]

b) Express the matrix $\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ as a sum of a symmetric and skew-symmetric matrix. [2]

c) Without expanding the determinant, prove that:

$$\begin{vmatrix} b+c & b & c \\ c+a & c & a \\ a+b & a & b \end{vmatrix} = 0$$ [2]

22. a) What is the geometrical meaning of derivative?

If $g(x) = \sqrt{x}$ and $h(x) = 5x - 1$, then find the composite function $(goh)(x)$. [1+1]

b) Find from first principles, the derivative of $f(x) = (goh)(x)$ [3]

c) Evaluate: $\lim_{x \to y} \left(x - \sqrt{x^2 + x}\right)$ [3]

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